

Project-Based CST&AI Foundations Syllabus

Project-Based CST&AI Foundations, Fall 2026, 3 Credits

Course Code / Section: 44511105 | Language of Instruction: English

Total Contact Hours: 88 (Lecture 8; Seminar 12; Vibe Coding 20; Projects 48)

Date Range, Meeting Time, and Classroom: TBA

Instructor Information

Instructors	Email	Office Hours & Location
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General Information

Course Description

Project-Based CST&AI Foundations is a required foundational course for first-year students in computing-related programs. It replaces a conventional introduction to computer science and artificial intelligence with a practice-centered course aligned with the institute's emphasis on strong capability and authentic engineering experience.

The course consists of four integrated modules: lectures, Seminar, Vibe Coding, and AI-assisted real-world projects. Its focus shifts from syntax memorization and isolated code writing toward computation and abstraction, engineering thinking, product thinking, teamwork, and AI-native software development.

The course is designed around three goals: to stimulate interest in computer science and AI engineering; to deepen industry-education integration through “real problems, real solutions, and real demonstrations”; and to explore how AI can support project-based learning through a layered learning pattern of “action-knowledge-action.”

Course Registration Details

This is a 3-credit required professional foundational course for students in Computer Science and Technology and Intelligent Medical Engineering (Brain-Computer Interface track). It includes 8 hours of lectures and 80 hours of project-oriented learning activities.

Pre- &/or Co-Requisites

Prerequisite: None. Co-requisite: Computer Programming.

Course Materials

Course Texts and References

- Instructor-developed course materials.
- [Vibe Coding Fundamentals: From Idea to First Project](#)
- [The AI-Native SDLC Playbook](#) from Claude Academy

Course Website and Classroom Management Tools

The designated learning management system will distribute slides, project briefs, seminar materials, milestone requirements, announcements, grades, and supplementary resources. Students are responsible for checking course announcements and official university email regularly.

Course Software

- An instructor-approved AI coding environment or AI-enabled IDE.
- Git-based version control for individual process records, team collaboration, and evidence of iterative development.
- Testing, debugging, documentation, and collaboration tools specified for each activity or project.
- LLM APIs, Tool Use, Function Calling, and Model Context Protocol (MCP) components when required by the activity.

Course Objectives

- Course Objective 1 - AI Foundations: Understand basic concepts, capability boundaries, and typical applications of AI and large language models, including the AI amplifier effect, prompt design, code review, and engineering constraints.
- Course Objective 2 - Computation, Abstraction, and Product Thinking: Understand computation and abstraction, complexity, layering, modularity, reuse, MVP design, and iterative product development.
- Course Objective 3 - Software Engineering Foundations: Understand problem decomposition, modular design, interface-implementation separation, data and state management, front-end/back-end collaboration, testing, debugging, and iteration.
- Course Objective 4 - AI-Assisted Development: Use AI programming tools responsibly and apply Prompt Engineering, Tool Use, Function Calling, MCP, context management, and Agentic Engineering to construct and verify prototypes.
- Course Objective 5 - Real-World Project Delivery: Complete the full process of requirements research, MVP specification, implementation, testing, real-user validation, version management, demonstration, delivery, and structured reflection.
- Course Objective 6 - Professional Responsibility: Develop awareness of AI ethics, data security, copyright compliance, engineering responsibility, and teamwork, and follow professional software-engineering norms.

Learning Outcomes

By the end of the course, a successful student should be able to:

- Explain the roles and limitations of AI and large language models in software development and distinguish plausible output from verified evidence.
- Translate an open-ended idea into a clearly stated problem, a decomposed set of requirements, an MVP specification, and an iterative project plan.
- Apply abstraction, modularity, interface design, state management, testing, debugging, and version control in a small software system.
- Use natural-language interaction and AI development tools to produce a working prototype while preserving prompts, decisions, modifications, and verification evidence.
- Work in a team to deliver a runnable, testable, demonstrable project that has received feedback from real users or relevant stakeholders.
- Review AI-generated code, identify defects or inappropriate assumptions, and revise the result based on correctness, security, maintainability, and engineering constraints.
- Document data sources, external code, AI assistance, licenses, and team contributions in a transparent and auditable manner.

Course Organization

The course includes 8 hours of lectures, 12 hours of Seminars, 20 hours of Vibe Coding, and 48 hours of AI-assisted real-world projects. Student practice is the primary learning mode; instructor mini-lectures, enterprise interaction, code review, user testing, oral defense, and structured reflection support the project work.

Lecture Modules (8 Hours)

Unit	Topic	Core Focus	Hours
1	The Nature of Computation and Abstraction	Computational models, complexity, layered abstraction, core CS areas, and foundational programming concepts.	2
2	Engineering Thinking	Problem formulation and decomposition; abstraction, modularity, reuse; software engineering; bugs and debugging.	2
3	Concepts in AI and Large Language Models	Intuition for LLMs, capability boundaries, hallucination, critical judgment, AI ethics, and development environments.	2
4	From Idea to Product	Product thinking, MVP, iteration, AI-native development, project narrative, and the real-world project roadmap.	2

Seminar Program (12 Hours)

Eight 1.5-hour Seminar sessions are held approximately every two weeks. Seminars introduce enterprise talent needs and authentic engineering problems, support student-enterprise dialogue, and contribute to project review and evaluation.

Seminar Expectations

- Prepare by reviewing the announced topic, enterprise context, or project materials.
- Participate in questions, discussion, roundtables, and project critique.

- Communicate requirements, constraints, progress, risks, and deliverables clearly to enterprise mentors and reviewers.
- Use seminar feedback to revise project scope, MVP specifications, implementation decisions, or delivery plans.

Vibe Coding (20 Hours)

Vibe Coding is scheduled primarily during Weeks 1-5. Instructors assign a topic and students use natural-language interaction with AI to create a rapid prototype. Memorization of syntax is not the primary instructional goal; after the initial stage, the course introduces problem decomposition, context management, and verification methods.

Core Topics

- AI-assisted development of a personal project from initial idea to working prototype.
- Iterative improvement of content, structure, design, and functionality.
- Publishing a working project and connecting it with appropriate supporting services.
- Collecting feedback, evaluating results, and making evidence-based improvements.

Required Process Evidence

- Representative records of AI-assisted development and key project decisions.
- Evidence of major iterations, version changes, and improvements.
- A runnable or publicly accessible prototype with basic testing and feedback evidence.

AI-Assisted Real-World Projects (48 Hours)

The second stage, primarily Weeks 6-16, uses AI-assisted programming to solve authentic problems. Project topics come from enterprise partners or student proposals approved by the course team. Students work through requirements, MVP design, implementation, testing, user validation, iteration, demonstration, and reflection.

Project Workflow and Required Deliverables

Stage	Primary Work	Required Evidence / Deliverable
1. Project Initiation	Topic selection, stakeholder communication, problem framing, team roles, and scope control.	Project proposal and MVP Product Requirements / Specification document.
2. MVP Sprint	Architecture, module boundaries, initial implementation, version control, testing, and debugging.	Runnable MVP with Git history and test evidence.
3. User Validation	Real-user or stakeholder trials, observation, feedback collection, and mid-project diagnosis.	User experience / feedback report and prioritized revision plan.
4. Iterative Development	Feature completion, defect correction, integration, robustness improvement, and Beta release.	Functionally complete Beta version and updated documentation.
5. Demo Day	Final integration, demonstration narrative, evidence packaging, presentation, and review.	Demo Day materials, live demonstration, and review responses.
6. Reflection and Archiving	Structured retrospective, AI audit, contribution record, and compliant release or internal archive.	Course summary report, AI collaboration record, and code release/archiving package.

Course Requirements & Grading

- The course is evaluated through weighted continuous assessment and recorded on a Pass/Fail basis.
- A total weighted score of at least 60 is required for Pass.
- In addition, the Projects component must be passed, and no serious academic-integrity or AI-compliance violation may occur.
- Process traceability is required. Relevant prompts, AI outputs, human modifications, tests, Git history, user feedback, and contribution records may be used to verify authorship and understanding.

Assessment Components

Assessment Component	Weight
Class Performance	10%
Lectures	10%
Seminar	10%
Vibe Coding	20%
Projects	40%
AI Compliance	10%
Total	100%

Evaluation Bands

Result	Weighted Score / Additional Conditions
Pass	60-100%, Projects passed, and no serious academic-integrity or AI-compliance violation.
Fail	Below 60%, or Projects not passed, or a disqualifying integrity/compliance violation.

Assessment Standards

The detailed rubrics, performance levels, evidence requirements, and component-specific pass/fail rules are as follows:

Class Performance (10%)

The course will be fully in-person. Attendance and classroom participation will be 10% of the overall grade. Class presence and participation points are given to encourage your active class participation and discussion. You will be rewarded with a perfect score if you frequently come to class and actively contribute to the class discussion during lectures.

Presence: Although it is not required, most students send their professor a brief e-mail to explain their absence in advance. Students who repeatedly arrive late to the lecture or recitation will have their Class Participation grade lowered. Please sign the attendance sheet when you come to the class. You will lose 1 point for each unexcused absence, up to a maximum of 5 points. Any false signatures will result in zero participation grades for all parties involved.

Participation: Although not explicitly graded, you will be evaluated on the QUALITY of your contributions and insights. Quality comments possess one or more of the following properties: Offers a different and unique, but relevant, perspective; Contributes to moving the discussion and analysis forward; Builds on other comments; Transcends the “I feel” syndrome. That is, it includes some evidence, argumentation, or recognition of inherent tradeoffs. In other words, the comment demonstrates some reflective thinking. I will use our assessment of your participation to manage borderline grades. While your participation grade is subjective, it will not be random or arbitrary. And, clearly, more frequent quality comments are better than less frequent quality comments.

Lectures (10%)

You will be assigned reading and video-watching materials as homework. As a general guideline, for every 1 hour of lecture, you are expected to spend at least 2 hours studying at home. Accordingly, I will give you quizzes or tests to check whether you have completed the assigned readings and watched the videos.

Seminar (10%)

After each seminar, students are required to complete an AI-based academic inquiry exercise:

1. Students raise 2-5 academic or research-oriented questions based on the seminar content.
2. Students conduct multi-round interactive discussions, verification, and critical questioning with AI tools.
3. The complete question-and-answer dialogue process must be fully submitted as coursework records.

Grading Rubric

1. Question Quality (4 points): Innovative, technical, mechanism-focused, or research-gap-oriented questions are highly scored. Superficial common-knowledge questions receive low scores.
2. AI Interaction Depth (4 points): Multiple rounds of questioning, verification, correction, and extended discussion are required. Single-turn dialogue is not acceptable.
3. Completeness & Standardization (2 points): Clear structure, topic relevance, and complete learning process records.

Vibe Coding (20%)

The vibe coding component accounts for 20% of your final grade, distributed across three deliverables. The largest portion, worth 10%, covers the personal homepage developed across Phases 1 through 4. An additional 5% is allocated to a final extension of that homepage, which requires either connecting a Digital Twin or knowledge base, or implementing one significant new feature of your choice. The remaining 5% comes from an independent mini project, where you will build a website or small application unrelated to the homepage and demonstrate that you can transfer the same iterative development approach to a new context.

Projects (40%)

You are encouraged to complete the project in groups, and you are free to form your own teams. Each group may have a maximum of 5 members. Your project grade will be calculated as follows: 70% from the Team Deliverable Grade, and 30% from the Individual Performance Score. The Team Deliverable Grade reflects the quality of your group's work across all project stages. The Individual Performance Score is based on two components: your AI usage analysis and your Git contribution history.

Stage 1: Project Initiation (6 points) - Your proposal must clearly define the problem, user needs, and success metrics, with evidence of stakeholder communication and realistic scope. The MVP PRD should include user stories, prioritized features, and technical assumptions.

Stage 2: MVP Sprint (8 points) - Your runnable MVP must demonstrate a clear architecture, regular Git commits with meaningful messages, and test or debugging evidence covering critical paths. The deliverable is the working prototype with version history and test documentation.

Stage 3: User Validation (6 points) - Conduct structured user trials, collect feedback through multiple channels, and produce a diagnosis with a prioritized revision plan. The feedback report should include raw data and clear next steps.

Stage 4: Iterative Development (8 points) - Complete all planned features, resolve critical defects from validation, and improve robustness. Submit a functional Beta version with updated documentation and evidence of regression testing.

Stage 5: Demo Day (6 points) - Present a clear narrative connecting user needs to technical outcomes, deliver a live demo that works, and submit polished materials (slides, script, review forms) with organized process evidence.

Stage 6: Reflection and Archiving (6 points) - Provide an honest retrospective, a complete AI interaction audit with critical evaluation, a fair team contribution record, and a well-organized code archive or release package.

AI Compliance (10%)

This component evaluates how you engage with AI tools throughout the project. You are encouraged to use AI for assistance, but you must document your interactions (via a submitted AI log), demonstrate that you critically evaluated AI suggestions, and show that you understand any AI-generated code or content in your project. Over-reliance on AI without verification, unclear attribution, or sharing sensitive data with AI tools will result in point deductions. The goal is to use AI as a collaborator—not a replacement—for your thinking.

Assessment Criteria

Your AI Compliance grade (10 points total) is based on the following five criteria, each worth 2 points:

1. **AI Usage Log:** Submit a complete, timestamped log of your key AI interactions (prompts and responses) that shows a clear progression from initial queries to refined outputs.
2. **Critical Evaluation:** Demonstrate that you questioned, verified, or corrected AI suggestions. Provide evidence of multiple rounds of refinement or debugging informed by your own judgment.
3. **Understanding of AI-Generated Content:** You must be able to explain AI-generated code or content in your own words, including its purpose, trade-offs, and limitations. This will be verified during Demo Day Q&A or in your reflection report.
4. **Appropriate Use & Attribution:** Use AI for assistance (e.g., boilerplate, debugging, refactoring) rather than for core conceptual work. Clearly cite or flag AI-generated sections in code comments or documentation.
5. **Ethical & Privacy Compliance:** Do not share sensitive data (e.g., credentials, personal information, proprietary code) with AI tools. Exercise judgment about what is appropriate to input into public or third-party AI systems.

Grading Philosophy

This is not a test of your AI tool proficiency. It is an assessment of your critical thinking, intellectual honesty, and ability to use AI as a thought partner rather than a substitute for your own reasoning. Over-reliance on AI without verification, unclear attribution, or failure to understand AI-generated content will result in point deductions. Conversely, thoughtful, documented, and ethically sound AI use will be rewarded.

Assignment Deadlines and Policies

Milestone Submission

- Project proposals, MVP specifications, prototypes, user-feedback records, Beta versions, demonstration materials, and reflection/archiving packages must be submitted through the designated platform by the announced deadlines.
- Students are responsible for confirming that files, repositories, links, demonstrations, and access permissions work before submission.

Late and Make-up Work Policy

- There are No-Makeups or Grade Replacements for homeworks.
- If you miss any assignment without a valid excuse, then you will receive a 0.
- Institute approved absences are rare and do qualify for an exception.
- Institute approved absences must be made in advance of the assessment unless it is physically impossible to do so.
- *Documentation* must be provided to the Dean of Students (NOT your professor) on letterhead with the signature of a physician, supervisor, or other appropriate official. Additionally, the documentation must encompass the date(s) of any assignment for which you are requesting an exception.
- The final decision regarding any absence is made solely at the discretion of your professor.
- Events such as vacations, parents set up travel plans, weddings, graduations, errands, work conflicts, sleeping through your alarm, alarm malfunction, forgetting to submit, forgetting the date or time of an exam, or not being aware of the assignment are all NOT valid excuses.
- If your absence is approved, then the professor will adjust the specific assignment(s) as they deem appropriate.

Grade Disputes / Regrades

Questions about recorded grades should be raised promptly and professionally through the process announced by the instructor. A regrade request must identify the specific evidence or requirement at issue. Reassessment may include review of the complete submission, process records, repository history, demonstration, and oral explanation.

Course Policies, Expectations & Guidelines

Communication Policy

- Use official university email and the designated course platform for course communication.
- Questions that contain private academic, grading, personal, enterprise, or project-sensitive information should be sent privately rather than posted publicly.
- Team members are expected to communicate progress, blockers, interface changes, and delivery risks in a timely and professional manner.
- Students should review course announcements and official email frequently enough to meet time-sensitive project and seminar requirements.

Student Conduct and (N)etiquette

- Read existing messages and project records before posting a repeated question or changing an agreed decision.
- Use professional, respectful, and evidence-based language in classroom, online, team, enterprise, and peer-review interactions.
- Do not disclose another person's private information or an enterprise partner's non-public information without permission.
- Keep repositories, shared folders, issue trackers, and communication channels organized and appropriately permissioned.
- Do not distribute malicious, offensive, deceptive, or unrelated content through course systems.

University Use of Electronic Email

The university-assigned student email account is an official means of communication. Students are responsible for information sent to that account and for maintaining reliable access to course announcements, seminar notices, feedback, and project deadlines.

Accommodations for Students with Disabilities

Students who require disability-related accommodations should contact the appropriate university office as early as possible and provide the instructor with the official accommodation notice. The course team will implement approved arrangements while preserving essential learning outcomes and authentic assessment.

Student-Faculty Expectations Agreement

The course depends on mutual respect, preparation, responsibility, transparency, and constructive feedback. Students may expect clear requirements, meaningful project guidance, and consistent application of course policies; the instructor expects honest work, sustained participation, professional teamwork, and responsible use of AI.

Student Use of Mobile Devices in the Classroom

Laptops and other devices may be used for course materials, development, testing, teamwork, note-taking, and approved AI interaction. Sounds and unrelated visual content should be disabled so that devices do not disrupt other students, guests, or enterprise partners.

Plagiarism & Academic Integrity

Students are expected to meet the highest standards of academic integrity. Submitted work must accurately represent the student's and team's own contributions. Permitted external assistance, open-source material, documentation, enterprise input, and generative-AI assistance must be acknowledged in the form required by the course.

Suspected plagiarism, cheating, fabrication of process evidence, unauthorized collaboration, or misrepresentation of authorship may be referred to the appropriate university process. Work that cannot be authenticated, explained, or demonstrated may receive no credit.

The Following Are Not Allowed

- Copying or lightly modifying another team's proposal, code, tests, documentation, presentation, user-feedback record, Git history, or AI collaboration record.

- Submitting material that is wholly or substantially identical to another person's work without authorization and appropriate attribution.
- Publishing restricted project briefs, enterprise data, assessment materials, hidden tests, credentials, or non-public solutions in a public repository.
- Allowing another person to complete or explain an individual assessment under your name, or misrepresenting the distribution of team contributions.
- Fabricating real-user validation, enterprise feedback, AI audits, data provenance, license compliance, or security testing.

Collaboration & Group Work

Collaboration is central to this course. Teams may share code and documents within the boundaries of their own approved project and course instructions. Cross-team discussion of general concepts is encouraged, but sharing another team's implementation, evidence, or deliverables is not permitted unless explicitly authorized.

Each team must maintain transparent task allocation and integration records. Every member is responsible for understanding the complete delivered system and for honestly reporting individual and shared contributions.

Copyright

Course slides, project briefs, assessment materials, partner information, review comments, and test resources are provided for educational use and may not be republished or distributed without permission. Third-party code, data, models, media, documentation, and AI services remain subject to their respective licenses and terms.

Subject to Change Statement

The syllabus, project milestones, seminar sequence, tools, and detailed schedule may be revised to improve learning or respond to enterprise, technical, or operational needs. Changes will be communicated through the learning management system and official email. Students are responsible for staying current.

Important Dates (all dates are tentative and subject to change)

Milestone	Date
First Class Day	September 1 st , 2026
Vibe Coding Prototype Review	September 29 th , 2026
Project Proposal and MVP Specification	October 13 th , 2026
Runnable MVP Review	October 27 th , 2026
User-Feedback / Mid-Project Review	November 10 th , 2026
Beta Version Review	December 1 st , 2026
Demo Day	December 8 th , 2026
Final Reflection and Project Archive	December 15 th , 2026